

HEALTHCONNECT EXPERIENCE LAB

Project Title: Healthconnect Clinic Advanced Analytics & Decision Support Report

Program: AnalystLab Africa (Data Analytics Track)

Week 6

Executive Summary

Week 6 advanced the HealthConnect Data Analytics work from the Week 5 baseline dashboard into deeper, segment-level analysis and decision support. The analysis focused on whether the strongest Week 5 patterns remained visible when previous no-show history, distance, reminder status, appointment type and age were examined together.

The Week 6 dashboard retained the Week 5 pages as a baseline and added two analytical pages: Advanced No-Show Analysis and Reminder & Risk Segmentation. The analysis validated the overall clinic no-show rate at 48.5% and identified previous no-show history and distance as priority risk areas for further testing. Reminder status also showed an observed difference in no-show rates, supporting further testing rather than a causal claim.

A documented Data Analytics → Project Management integration was completed during Week 6. The Data Analytics output was received and reviewed by Project Management and used to refine the integration/dependency view and Week 7 testing preparation.

Week 5 → Week 6 Transition

Week 5 established the HealthConnect appointment-attendance baseline through data preparation, exploratory analysis, key performance indicators and an initial three-page Power BI dashboard. The Week 5 dashboard provided the baseline for no-show, cancellation, reminder coverage and waiting-time performance. The Week 5 analytical foundation was preserved and extended through derived segmentation, cross-variable analysis and KPI validation.

Week 5 baseline	Week 6 advancement
Overall no-show rate and outcome profile	Validated baseline and examined high-risk segments
Reminder channel and coverage patterns	Examined reminder status across appointment type, time and distance
Appointment-type patterns	Examined appointment type together with previous no-show history and reminder channel
Previous no-show history	Grouped history into 0, 1–2 and 3+ and examined interactions with other factors
Distance pattern	Examined distance together with previous no-show history and reminder status
Three-page dashboard	Retained baseline pages and added two advanced analytical pages

Week 6 Analytical Objectives

- Investigate the strongest Week 5 no-show patterns at a deeper segment level.
- Determine whether key patterns remain meaningful when patient and appointment characteristics are examined together.
- Validate important Week 5 KPIs against the Week 6 analytical model.
- Identify high-priority risk segments for business action and Week 7 testing.
- Translate analytical findings into practical HealthConnect actions without overstating association as causation.
- Provide evidence-based findings and limitations that can support cross-track project coordination.

Advanced Analysis

1. Previous No-Show History

Previous no-show history was grouped into three practical segments: 0 previous no-shows, 1–2 previous no-shows and 3+ previous no-shows. The observed no-show rates were 43.5%, 54.8% and 68.8%, respectively.

This shows a strong increasing pattern and makes previous no-show history a priority segment for Week 7 testing. The 3+ group should nevertheless be interpreted with caution if its underlying appointment count is small.

2. Distance from the Clinic

The 21+ km group recorded a 57.8% no-show rate compared with the overall clinic rate of 48.5%. The Week 6 cross-analysis also examined distance alongside previous no-show history and reminder status.

Distance should be treated as an associated risk factor requiring further investigation rather than proof that travel distance causes missed appointments.

3. Reminder Status and Coverage

Reminder coverage remained 72.7%. Appointments without reminders recorded a 51.4% observed no-show rate, while appointments with reminders recorded a 47.4% observed no-show rate, giving a 4.0 percentage-point observed gap.

The finding supports further testing of reminder coverage and escalation strategies. It does not establish that reminders alone caused the lower no-show rate.

4. Appointment Type

Follow-up appointments had the highest observed no-show rate among appointment types at 51.2%, followed by Diagnostic Test at 49.7%, Specialist Consultation at 47.4% and General Consultation at 46.6%.

Appointment type therefore remains a relevant segmentation variable, but the observed differences are less pronounced than the pattern associated with previous no-show history.

5. Segment Interactions

The Week 6 dashboard examined combinations of risk factors rather than relying only on single-variable comparisons. These included appointment type × previous no-show history, distance × previous no-show history, distance × reminder status, age group × previous no-show history, appointment type × reminder channel, and appointment time × reminder status.

The purpose of these views was to identify whether risk patterns were consistent across segments and to identify areas suitable for controlled testing.

Validated Findings

Finding	Evidence	Interpretation / Validation
Overall no-show remains high	48.5% overall no-show rate	Week 5 baseline was reproduced in Week 6.
Previous no-show history is a strong risk signal	43.5% for 0; 54.8% for 1–2; 68.8% for 3+	Clear increasing observed pattern; priority for further testing.
Distance is relevant	21+ km = 57.8% vs 48.5% overall	Elevated observed rate; investigate accessibility/travel-related barriers.
Reminder status shows an observed difference	51.4% no reminder vs 47.4% reminder; 4.0 pp gap	Supports reminder testing; does not prove causation.
Follow-up appointments have the highest appointment-type rate	Follow-up = 51.2%	Relevant segment for targeted engagement/testing.

Refined KPIs

The Week 6 KPI set preserves the four strong baseline KPIs and adds diagnostic measures for deeper decision support.

KPI / Measure	Week 6 value	Purpose
No-Show Rate	48.5%	Primary attendance-risk KPI; validates Week 5 baseline.
Cancellation Rate	5.3%	Tracks scheduled appointments lost through cancellation.
Reminder Coverage Rate	72.7%	Measures the proportion of appointments receiving reminders.
Average Waiting Time	24.19 minutes	Monitors patient-support/operational waiting experience.
Repeat No-Show Rate	55.4%	Observed no-show rate among appointments with previous no-show history.
No-Reminder No-Show Rate	51.4%	Observed no-show rate for appointments without a reminder.
Reminder No-Show Rate	47.4%	Observed no-show rate for appointments with a reminder.
Reminder No-Show Gap	4.0 percentage points	Observed difference between no-reminder and reminder groups.

Business Insights and Implications

Insight	Business implication
Patients with previous no-show history have higher observed no-show rates.	A general reminder approach may not be sufficient for repeat-risk patients.
The 21+ km group has an elevated observed no-show rate.	Accessibility or travel-related barriers may affect attendance.
No-reminder appointments have a higher observed no-show rate than reminder appointments.	Reminder coverage is a practical intervention area.
Follow-up appointments have the highest appointment-type no-show rate.	Follow-up patients may need additional engagement.
Booking lead-time data has quality concerns.	Using an unreliable variable could weaken downstream modelling or decisions.

Recommendations

1. Prioritize previous no-show history for risk-based attendance interventions and testing.
2. Investigate distance-related attendance barriers, especially for the 21+ km segment.
3. Test reminder coverage and escalation strategies rather than assuming a single reminder channel will work for every segment.
4. Monitor follow-up appointments as a higher-risk appointment segment.
5. Improve the quality and definition of booking lead-time data before using it for predictive or downstream technical work.
6. Use Week 7 testing to determine whether the observed risk patterns remain stable and whether targeted interventions produce measurable improvement.

Cross-Track Contribution / Integration: Data Analytics ⇄ Project Management

Integration Objective

The purpose of the Data Analytics ⇄ Project Management integration was to connect appointment attendance findings with project coordination, dependency management and Week 7 testing preparation.

What Project Management Gave Data Analytics

- Project integration context and the role of Project Management as the coordination layer across technical workstreams.
- Current dependency, risk, issue and priority considerations relevant to the next project stage.
- Week 7 testing/refinement focus, including the need to verify workstream outputs and integration readiness.
- A framework for distinguishing planned integration pathways from completed integrations supported by actual evidence.

What Data Analytics Gave Project Management

- The Week 5 HealthConnect appointment attendance/no-show analysis.
- Key no-show KPIs and findings.
- Relevant factors and segments requiring further investigation.
- Data-quality limitations and remaining analytical work.
- Analytical priorities that could inform Week 7 testing and refinement.

What Was Actually Exchanged

The Data Analytics output was provided to Project Management and reviewed against the HealthConnect objective. Project Management supplied the integration context, dependency considerations, priorities and Week 7 testing/refinement requirements. This created a two-way exchange rather than communication alone.

What Changed as a Result

The analytical findings were incorporated into the Project Management integration/dependency view and Week 7 preparation. Previous no-show history and other validated risk areas were carried forward as analytical considerations for testing, while remaining analysis and limitations were retained as open work rather than being treated as completed.

How the Output Was Used

The Data Analytics output was used to refine PM coordination and testing preparation. The integration record treats this as a completed Data Analytics → PM integration point because an actual output was received, reviewed and used to change or strengthen project controls. No unverified claim is made that Data Science, ML Engineering or Generative AI has already used the findings.

Week 7 Testing Requirements

Before Testing

- Confirm required workstream deliverables and supporting evidence are available.
- Review critical integration dependencies and close, accept or escalate them.
- Confirm testing criteria, expected outputs and responsibilities.
- Verify remaining technical workstream readiness.
- Review the data-quality status of variables intended for downstream modelling.

During Testing

- Test enhanced reminder strategies for patients with previous no-show history.
- Test targeted support for patients living 21+ km from the clinic.
- Evaluate reminder coverage and channel performance across relevant appointment segments.
- Monitor follow-up appointments as a higher-risk segment.
- Track test issues, severity, ownership, impact and resolution status.

After Testing

- Determine whether identified issues were resolved, deferred or accepted.
- Recalculate key KPIs and compare results with the Week 6 baseline.
- Update risks and dependencies based on test outcomes.
- Document limitations and required refinements.
- Assess readiness for broader integration and final project activities.

Limitations and Assumptions

- The analysis identifies associations and observed differences; it does not establish causation.
- Small high-history segments, particularly the 3+ previous no-show group, should be interpreted cautiously.
- Booking lead-time information was not treated as a validated Week 6 finding because of previously identified data-quality concerns.
- Waiting-time differences were not treated as a primary no-show driver because observed outcome averages were very similar.
- Technical outputs from Data Science, ML Engineering and Generative AI were not independently verified in the PM evidence set.
- The completed cross-track integration is specifically Data Analytics → Project Management; other technical pathways remain pending evidence.
- The Week 6 analysis uses the cleaned HealthConnect dataset established for the project; original data should remain preserved separately from processed/derived files.

Week 6 Conclusion

Week 6 moved the HealthConnect Data Analytics contribution from baseline reporting to deeper analytical decision support. The overall 48.5% no-show rate was validated, while previous no-show history, distance, reminder status and appointment type were examined across multiple segments. Previous no-show history and the 21+ km distance group emerged as priority areas for further testing, while reminder coverage remains a practical intervention area.

The completed Data Analytics ⇌ Project Management integration connected these findings to project coordination and Week 7 preparation. The next stage is testing and refinement: verify remaining technical outputs, test targeted interventions, monitor unresolved dependencies, and validate whether the observed patterns remain consistent in an end-to-end setting.